Phrase-O-Matic

How it works

In a nutshell, the program makes three lists of words, then randomly picks one word from each of the three lists, and prints out the result. Don't worry if you don't understand *exactly* what's happening in each line. For goodness sake, you've got the whole book ahead of you, so relax. This is just a quick look from a 30,000-foot outside-the-box targeted leveraged paradigm.

1. The first step is to create three String arrays—the containers that will hold all the words. Declaring and creating an array is easy; here's a small one:

```
String[] pets = {"Fido", "Zeus", "Bin"};
```

Each word is in quotes (as all good Strings must be) and separated by commas.

2. For each of the three lists (arrays), the goal is to pick a random word, so we have to know how many words are in each list. If there are 14 words in a list, then we need a random number between 0 and 13 (Java arrays are zero-based, so the first word is at position 0, the second word position 1, and the last word is position 13 in a 14-element array). Quite handily, a Java array is more than happy to tell you its length. You just have to ask. In the pets array, we'd say:

```
int x = pets.length;
```

and **x** would now hold the value 3.

3. We need three random numbers. Java ships out of the box with several ways to generate random numbers, including java.util.Random (we will see later why this class name is prefixed with java.util). The **nextInt()** method returns a random number between 0 and some-number-we-give-it, *not including* the number that we give it. So we'll give it the number of elements (the array length) in the list we're using. Then we assign each result to a new variable. We could just as easily have asked for a random number between 0 and 5, not including 5:

```
int x = randomGenerator.nextInt(5);
```

4. Now we get to build the phrase, by picking a word from each of the three lists and smooshing them together (also inserting spaces between words). We use the "**+**" operator, which *concatenates* (we prefer the more technical *smooshes*) the String objects together. To get an element from an array, you give the array the index number (position) of the thing you want by using:

```
String s = pets[0]; // s is now the String "Fido"
s = s + " " + "is a dog"; // s is now "Fido is a dog"
```

5. Finally, we print the phrase to the command line and...voilà! *We're in marketing*.

what we need here is a...

extensible microservices pipeline

opinionated loosely coupled REST API

agent-based microservices library

AI-enabled service oriented orientation

agnostic pub-sub DSL

functional IoT perspective